Differential Memory Decay: A Continuous-Time Model of Forgetting and Semantic Drift

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Abstract

Forgetting is not a defect of memory systems but a prerequisite for long-term stability, generalization, and adaptive intelligence. Biological memory relies on multiple interacting mechanisms of decay, including synaptic weakening, semantic drift, and structural pruning to prevent interference, regulate representational capacity, and maintain coherence over time. In contrast, contemporary artificial memory systems used in large language models and agentic frameworks typically accumulate information indefinitely, lacking principled mathematical formulations for controlled forgetting. This discrepancy leads to semantic overload, unbounded memory growth, and instability in long-horizon reasoning.

In this work, we develop a continuous-time mathematical framework for *differential memory decay*, grounded in both computational neuroscience and energy-based dynamics. We formalize several distinct classes of decay operators, including linear forgetting at multiple timescales, category specific decay matrices, competition-driven normalization, and graph-Laplacian diffusion that induces semantic smoothing and drift. We further show that replay, modeled as energy descent in the absence of external input, acts as an anti-decay mechanism that selectively stabilizes long-term attractors while allowing transient information to fade.

Through analytical results and synthetic experiments, we demonstrate how the interaction between decay, diffusion, and replay yields memory representations that remain compact, stable, and semantically coherent over extended time horizons. Unlike prior work that focuses on storage or retrieval mechanisms, this paper isolates forgetting as an independent and valuable computational primitive. The framework presented here is architecture-agnostic and provides a theoretical foundation for integrating biologically motivated forgetting into future artificial memory systems.

Contents

1 Introduction

Forgetting is a central component of biological intelligence. Far from being a flaw or a failure mode, the ability to forget enables organisms to maintain stable representations, prevent runaway interference, and adapt to continuously changing environments. Neuroscientific research has demonstrated that forgetting operates across multiple temporal and structural scales: rapid decay of neuronal activations occurs on the order of milliseconds; synaptic weakening unfolds over seconds to minutes; and structural processes such as dendritic pruning or synaptic downscaling occur over hours, days, or sleep cycles. These mechanisms ensure that memory remains flexible, metabolically efficient, and robust to noise while still supporting long-term stability [Hebb, 1949, Dayan and Abbott, 2001, Buzsáki, 2015].

Modern artificial systems, however, rarely incorporate principled mechanisms of forgetting. Large language models (LLMs), agent memory buffers, and vector-database retrieval systems typically operate as static stores: information accumulates monotonically without decaying unless manually erased. Over extended deployments, such systems exhibit symptoms analogous to biological interference: oversaturation from irrelevant fragments, semantic drift between stored patterns, loss of coherence in retrieval, and unbounded growth of memory. Because these systems lack intrinsic decay, the responsibility for regulating memory falls entirely on external heuristics such as pruning rules, fixed-size buffers, or ad-hoc summarization. These approaches are brittle and fail to reflect the rich dynamics that make biological memory both plastic and stable.

This paper argues that intentional, mathematically grounded forgetting is essential for long-lived artificial agents. Just as different biological memory systems operate at distinct temporal scales, different types of artificial memory, including episodic traces, semantic concepts, task states, and long-term preferences should decay at different rates. Short-lived memories must fade quickly to avoid interference; intermediate memories should drift unless reinforced; and stable conceptual structures should decay slowly but not remain static indefinitely. This requirement motivates a theory of *differential memory decay*, in which forgetting is not uniform but tailored to the functional role of each memory dimension.

To support this perspective, we introduce a continuous-time mathematical framework for decay and forgetting in artificial memory systems. Rather than treating decay as deletion or noise injection, we model forgetting as a family of operators acting on a memory vector $M(t)$, each motivated by well-understood principles in computational neuroscience and energy-based modeling. These operators include (i) linear exponential decay across multiple timescales, (ii) category-specific decay controlled by a diagonal decay matrix, (iii) global inhibition and competition-based normalization, (iv) graph Laplacian diffusion that induces semantic drift, and (v) replay as an anti-decay process that selectively stabilizes long-term attractors. Each operator is independent of any specific memory architecture, making the framework broadly applicable to systems based on embeddings, energy networks, Hopfield dynamics, or agent-centric memory stores.

Importantly, this paper isolates controlled forgetting as an independent computational primitive. We do not describe or rely on any proprietary memory architecture, associative recall mechanism, or graph rewiring procedure. Instead, we provide theoretical tools for understanding how forgetting should be *designed*, *parameterized*, and *analyzed* in artificial memory. This level of abstraction enables future systems to incorporate biologically inspired decay without requiring the full machinery of attractor networks or continuous-time substrates.

The contributions of this paper are threefold:

- We present a conceptual taxonomy of memory types and argue for distinct decay profiles for episodic, semantic, structural, and task-dependent memory dimensions.
- We derive a unified continuous-time formulation of decay using linear operators, competition terms, and graph Laplacian diffusion, providing analytic expressions and stability interpretations.
- We demonstrate through synthetic experiments how decay, drift, and replay interact to produce compact, stable, and semantically coherent memory trajectories over extended time horizons.

By treating forgetting not as a side effect but as a structurally necessary component of intelligence, this work lays the mathematical foundation for future artificial systems with stable, adaptive, and biologically grounded long-term memory.

2 Background and Motivation

Motivation

Artificial memory systems lack the rich temporal and structural dynamics that enable biological memory to remain stable, adaptive, and compact over extended time horizons. Most contemporary AI systems treat memory as a form of static storage: vector databases, replay buffers, key–value archives, or unbounded session logs. Such systems accumulate information monotonically unless explicitly cleared, which leads to issues of semantic clutter, interference, and loss of coherence in retrieval. The challenge is not merely that these systems remember too much, but that they lack the capacity to *forget appropriately.*

Biological memory, in contrast, is inherently dynamic. Neuronal activations decay rapidly; synaptic strengths weaken or strengthen in response to usage; and large-scale systems such as the hippocampus and neocortex cooperate to determine which memories are consolidated or discarded. The resulting memory system is not simply a store of data but a continuously evolving dynamical substrate shaped by decay, drift, and consolidation. This motivates the central goal of this work: to develop a mathematically grounded framework that models forgetting as a first-class computational mechanism, rather than an afterthought or heuristic.

2.1 Biological Foundations of Forgetting

Biological memory exhibits several forms of decay, each operating on different temporal and structural scales:

Fast synaptic and neuronal decay. Neuronal firing rates decay in milliseconds, and post-synaptic potentials vanish on short timescales. This ensures that transient signals do not accumulate into long-term structure unless reinforced.

Intermediate synaptic decay and homeostatic regulation. Synaptic strengths are constantly adjusted through long-term depression (LTD), synaptic scaling, and metaplasticity. These processes prevent uncontrolled synaptic growth and maintain stability in neural circuits [Dayan and Abbott, 2001].

Slow structural decay. Over longer timescales, unused synapses are pruned and dendritic branches reorganize. This morphological forgetting enables efficient resource allocation and long-term stability.

Sleep and replay. During sleep, neural circuits replay recent experiences, selectively consolidating some memories while allowing others to fade [Buzsáki, 2015]. Replay counterbalances decay and prevents catastrophic forgetting.

These mechanisms collectively preserve important memories while ensuring that irrelevant or redundant information fades naturally.

2.2 Limitations of Artificial Memory Systems

Modern artificial memory systems operate fundamentally differently:

- Memory stores grow without bound unless manually pruned.
- Vector embeddings remain static and do not decay over time.
- Retrieval quality degrades as irrelevant items accumulate.
- Summaries produced by LLMs exhibit drift and lose fine-grained information.
- There is no mechanism to regulate representational stability.

This mismatch limits the long-term reliability of agentic systems or persistent LLM deployments.

2.3 Decay in Classical Associative Memory Models

Hopfield networks [Hopfield, 1982] and their modern extensions [Ramsauer et al., 2020, Krotov and Hopfield, 2016] provide attractor-based frameworks for memory storage. However, classical models assume static weights and do not incorporate forgetting. While energy-based models provide interpretability and stability guarantees, they lack mechanisms to remove outdated information or manage growth over time.

This motivates the need for explicit decay operators that can be incorporated into or alongside associative recall models.

2.4 Graph-Theoretic Representations and Semantic Drift

In modern machine learning, relational structure is often encoded through graph embeddings or affinity matrices. These structures naturally support diffusion processes governed by the graph Laplacian [Chung, 1997]. Laplacian dynamics tend to smooth or average representations over local neighborhoods, leading to *semantic drift*. This drift can be interpreted as a slow form of forgetting, where sharp or high-frequency distinctions fade while broad conceptual clusters persist.

This connection between spectral graph theory and memory decay provides an analytic foundation for modeling semantic forgetting in artificial systems.

2.5 Motivation for a Unified Decay Framework

Taken together, insights from neuroscience, associative memory, and spectral graph theory suggest that decay is not a single process but a family of operators that act across multiple scales and representations. Artificial systems need analogous mechanisms in order to:

- prevent uncontrolled memory growth,
- maintain stability in long-lived representations,
- avoid interference between transient and persistent memories,
- adapt semantic structure in response to long-term usage,
- preserve only information that is reinforced or replayed.

This motivates the development of a continuous-time framework capable of describing multiple classes of decay, semantic diffusion, and replay-mediated stabilization. The remainder of this paper builds such a framework.

3 Types of Memory Decay

Overview

Forgetting in artificial memory systems is not a monolithic process. Biological systems demonstrate a rich hierarchy of decay mechanisms that operate on different timescales and affect different components of memory. Short-lived neuronal activations dissipate quickly, synaptic strengths weaken gradually, semantic associations drift over time, and structural connectivity evolves on the slowest timescale. A meaningful theory of forgetting for artificial systems must mirror this layered, heterogeneous organization.

This section develops a taxonomy of decay processes, motivated jointly by computational neuroscience and dynamical systems theory. We emphasize that decay rates must be *differentiated* rather than uniform, because different memory types serve different roles and demand different stability profiles.

We categorize decay into four primary classes:

1. **Fast decay**: rapid dissipation of short-term activity.
2. **Medium-scale decay**: slow drift and soft forgetting of intermediate memories.
3. **Slow decay**: long-term erosion of stable or semantic memories.
4. **Structural decay**: gradual modification of the underlying semantic or relational graph.

Each class is formalized using continuous-time operators and analyzed in the context of artificial memory systems.

3.1 Motivation for Differentiated Decay

A uniform decay rate across all memory dimensions is inappropriate. Short-lived activations must fade rapidly, while conceptual knowledge must remain stable over long durations. This mirrors the division of memory responsibilities across the hippocampus, prefrontal cortex, and neocortex in biological systems.

Using a single decay constant α cannot capture this diversity:

- Large α destroys long-term memories.
- Small α preserves irrelevant short-term noise.
- Wrongly assigned α values destabilize the memory system.

A differentiated decay architecture is therefore essential.

3.2 Fast Decay: Dissipation of Activation Traces

Fast decay models the instantaneous fading of neuronal activations, a process operating on milliseconds to seconds. In artificial systems, this corresponds to transient computation artifacts such as:

- short-lived working-memory vectors,
- chain-of-thought residuals,
- momentary attention patterns,
- ephemeral reasoning states.

Mathematically:

$$\frac{dM}{dt} = -\alpha_{\text{fast}} M, \qquad \alpha_{\text{fast}} \gg 1.$$

Closed-form solution:

$$M(t) = e^{-\alpha_{\text{fast}} t} M(0).$$

Fast decay stabilizes computation by preventing instantaneous signals from bleeding into long-term memory.

3.3 Medium-Term Decay: Soft Forgetting and Semantic Drift

Medium-term decay captures slow weakening of representations on timescales of seconds to minutes (biological) or hours to days (artificial).

This is where **semantic drift** emerges: representations become smoother, less noisy, and gradually move toward local averages.

The operator is:

$$\frac{dM}{dt} = -\alpha_{\text{mid}} M - \gamma L M, \qquad \alpha_{\text{mid}} \ll \alpha_{\text{fast}}.$$

Here:

- $-\alpha_{\text{mid}} M$ removes unused intermediate information,
- $-\gamma L M$ smooths M over the semantic graph, inducing conceptual drift.

This allows the memory system to slowly reorganize and abstract over time.

3.4 Slow Decay: Long-Term Forgetting of Stabilized Memories

Slow decay corresponds to months- or years-scale forgetting in biological systems. In artificial systems, it models extremely slow erosion of:

- stable semantic knowledge,
- identity-level preferences,
- deep user traits,
- well-consolidated attractors.

Operator:

$$\frac{dM}{dt} = -\alpha_{\text{slow}} M, \qquad \alpha_{\text{slow}} \ll \alpha_{\text{mid}}.$$

This form of forgetting is deliberate and gentle, avoiding catastrophic loss of knowledge while still allowing outdated long-term content to fade.

3.5 Category-Specific Decay via Diagonal Operators

To express differentiated decay across memory dimensions, we introduce:

$$A = \operatorname{diag}(\alpha_1, \ldots, \alpha_d), \quad \frac{dM}{dt} = -AM.$$

Each coordinate may correspond to:

- an episodic index,
- a semantic factor,
- a preference dimension,
- a structural latent variable.

Working-memory dimensions receive large α_i, semantic-centroid dimensions receive small α_i, etc.

3.6 Competition-Driven Forgetting

Competition models the biological role of inhibitory interneurons. Artificially, it ensures global normalization of memory magnitude:

$$\frac{dM}{dt} = -\delta(1^\top M)\, 1.$$

This implements:

- resource constraints,
- suppression of runaway activations,
- competition between co-active patterns.

It complements linear decay by regulating global activity.

3.7 Structural Decay: Evolution of Memory Topology

Structural decay is the slowest process, occurring over days to weeks. It refers to the evolution of the **memory graph** W rather than the memory vector M itself.

Though the full system is not described in this paper, the functional role is:

$$W_{ij}(t + \Delta t) = \mathcal{F}(W_{ij}(t), \|M_i - M_j\|, \text{usage statistics}, \text{drift signal}),$$

where $\mathcal{F}$ is a pruning-and-strengthening rule.

Structural decay:

- prunes weak or obsolete semantic edges,
- merges or splits clusters,
- reorganizes conceptual topology based on long-term patterns.

3.8 Unified Timescale Hierarchy

The decay hierarchy can be summarized as:

$$\alpha_{\text{fast}} \gg \alpha_{\text{mid}} \gg \alpha_{\text{slow}} \gg 0.$$

This ensures that:

- activations vanish quickly,
- intermediate memories drift unless reinforced,
- semantic memories remain stable,
- topology evolves only very slowly.

Decay Taxonomy Table

Decay Type	**Timescale**	**Mathematical Form**	**Functional Role**
Fast decay	ms–s	$-\alpha_{\text{fast}} M$	Clear short-lived activations; prevent contamination of long-term memory.
Medium decay	s–min	$-\alpha_{\text{mid}} M - \gamma L M$	Semantic drift; smoothing of noisy representations; early abstraction.
Slow decay	min–hr–days	$-\alpha_{\text{slow}} M$	Long-term forgetting; removal of unused or obsolete knowledge.
Structural decay	hrs–weeks	$W_{ij}(t) \to \mathcal{F}(W_{ij})$	Topology reorganization; pruning weak edges; long-term reshaping of semantic structure.

Table 1: Hierarchy of decay operators, their typical timescales, mathematical representations, and functional roles. Column widths are fixed to ensure proper formatting without overflow.

3.9 Summary

This taxonomy provides the conceptual foundation for differential memory decay. The next section develops the unified mathematical formulation, where these decay processes are expressed as linear and graph-based operators within a continuous-time dynamical system.

4 Mathematical Model of Decay

Overview

This section presents a unified mathematical framework for modeling memory decay as a continuous-time dynamical system. Rather than treating forgetting as deletion or noise injection, we formalize it using linear differential operators, spectral diffusion, and global inhibition. These operators act on the memory state $M(t) \in \mathbb{R}^d$, representing a collection of memory features, semantic coordinates, or latent factors.

4.1 Base Form: Linear Exponential Decay

The simplest form of forgetting is exponential decay:

$$\frac{dM}{dt} = -\alpha M,$$

where $\alpha > 0$ controls the rate of forgetting.

Solution. The closed-form solution is:

$$M(t) = e^{-\alpha t} M(0).$$

Stability. Since $\alpha > 0$, the system is exponentially stable:

$$\lim_{t \to \infty} M(t) = 0.$$

Fast decay corresponds to large α; slow decay corresponds to small α.

4.2 Differential Decay via a Diagonal Operator

To allow heterogeneous forgetting across memory dimensions, we generalize the scalar decay to a diagonal matrix:

$$A = \mathrm{diag}(\alpha_1, \alpha_2, \ldots, \alpha_d), \qquad \alpha_i > 0.$$

The ODE becomes:

$$\frac{dM}{dt} = -AM.$$

Interpretation. Each component $M_i(t)$ satisfies:

$$\frac{dM_i}{dt} = -\alpha_i M_i,$$

leading to:

$$M_i(t) = M_i(0) e^{-\alpha_i t}.$$

Functional significance. This expresses differentiated decay across:

- episodic dimensions (large α_i),
- semantic dimensions (small α_i),
- structural or slowly varying components (very small α_i).

4.3 Global Inhibitory Decay

Biological memory systems contain global inhibitory circuits that regulate overall activation levels. We formalize this as a competition term:

$$\frac{dM}{dt} = -\delta(1^\top M)\,1, \qquad \delta > 0.$$

Interpretation. If the sum of memory activations is positive, all components are suppressed; if it is negative, all components rise uniformly.

Stability. The term encourages:

- global normalization,
- suppression of runaway activations,
- competition between co-active states.

This form is widely used in computational neuroscience to model lateral inhibition.

4.4 Semantic Drift via Graph Laplacian Diffusion

To incorporate semantic structure, we model memory as evolving over a graph with adjacency matrix W and degree matrix D. The graph Laplacian is:

$$L = D - W.$$

Semantic drift is modeled by:

$$\frac{dM}{dt} = -\gamma L M, \qquad \gamma > 0.$$

Interpretation. The term $LM = DM - WM$ enforces smoothing:

- each component is pulled toward the weighted average of its neighbors,
- sharper or noisy memory differences fade,
- clusters become internally coherent.

4.5 Spectral Analysis of Diffusion

Let $L = U\Lambda U^\top$ be the eigen decomposition of the Laplacian. Then the solution is:

$$M(t) = Ue^{-\gamma\Lambda t}U^\top M(0).$$

Consequences.

- Eigenvectors with large eigenvalues decay quickly (high-frequency noise).
- Eigenvectors with small eigenvalues persist (semantic clusters).
- The steady-state solution lies in the kernel of L, i.e., a piecewise-constant semantic manifold.

This formalizes *semantic drift* as the loss of high-frequency distinctions.

4.6 Unified Decay ODE

Collecting the linear, type-specific, inhibitory, and graph-diffusion terms, we obtain the unified decay dynamics:

$$\frac{dM}{dt} = -AM \; - \; \gamma LM \; - \; \delta(1^\top M)1.$$

We call this the **core decay operator**:

$$\mathcal{D}(M) = -AM - \gamma LM - \delta(1^\top M)1.$$

Interpretation. $\mathcal{D}(M)$ defines a continuous-time forgetting process combining:

- local exponential decay,
- global competition,
- graph-based smoothing.

It is architecture-agnostic and can be embedded into any memory substrate.

4.7 Closed-Form Solution for the Linear System

Assuming A and L commute (true when $A = \alpha I$ or under mild conditions),

$$\frac{dM}{dt} = -(A + \gamma L)M$$

has the solution:

$$M(t) = e^{-(A+\gamma L)t} M(0).$$

Spectral decomposition. Let $A + \gamma L = U \Sigma U^\top$, then:

$$M(t) = U e^{-\Sigma t} U^\top M(0).$$

Each eigencomponent decays at its own combined rate.

4.8 Stability Analysis

Because:

$$A \succ 0, \quad L \succeq 0, \quad (1^\top M)1 \text{ is bounded},$$

the system is globally exponentially stable:

$$\lim_{t \to \infty} M(t) = \mathbf{0}.$$

The only trajectories that persist are those lying entirely in the nullspace of $A + \gamma L$, corresponding to slowly varying semantic manifolds.

4.9 Role in Artificial Memory Systems

The unified decay model supports:

- **capacity management**: unused memories fade automatically,
- **semantic coherence**: drift smooths spurious distinctions,
- **temporal abstraction**: short-term noise disappears faster than concepts,
- **continuous adaptation**: memory reshapes in response to usage patterns.

This ODE defines the mathematical backbone for principled forgetting in artificial memory.

4.10 Transition to Diffusion and Replay Dynamics

While the operator $\mathcal{D}(M)$ defines forgetting, stability of long-term attractors requires counteracting forces. The next section introduces *semantic diffusion*, and later we incorporate *replay* as a balancing process that strengthens meaningful memories while allowing irrelevant ones to dissipate.

5 Semantic Diffusion and Drift

Overview

Semantic drift refers to the gradual evolution of memory representations as they interact with related items in a structured space. In biological systems, drift arises through synaptic co-activation, cortical smoothing, and systems consolidation. In artificial systems, semantic drift is most naturally formalized using graph-based diffusion. This section introduces the mathematical foundations of semantic diffusion, its spectral interpretation, and its functional relevance for memory stability.

5.1 Memory as a Function on a Graph

We assume memory is organized over a semantic graph $\mathcal{G} = (V, E, W)$, where:

- V is the set of memory nodes,
- $W \in \mathbb{R}^{d \times d}$ is a symmetric affinity matrix,

- W_{ij} measures the semantic similarity between i and j.

The degree matrix is:

$$D_{ii} = \sum_j W_{ij}.$$

The graph Laplacian is:

$$L = D - W.$$

This operator plays a central role in diffusion dynamics, as it penalizes sharp variations across edges.

5.2 Laplacian Diffusion as Semantic Drift

We model semantic drift through the continuous-time diffusion equation:

$$\frac{dM}{dt} = -\gamma LM, \qquad \gamma > 0.$$

Interpretation. The term $LM = DM - WM$ performs:

- **local averaging**: each memory is pulled toward its neighbors,
- **noise suppression**: high-frequency variations decay rapidly,
- **cluster consolidation**: conceptually related memories converge.

This mirrors cortical smoothing mechanisms and ensures conceptual coherence in artificial systems.

5.3 Closed-Form Solution

If $L = U\Lambda U^\top$ is the eigen-decomposition of the Laplacian, then:

$$M(t) = Ue^{-\gamma\Lambda t}U^\top M(0).$$

Because Λ is diagonal with non-negative entries:

- high-eigenvalue modes (*high-frequency, noisy*) decay rapidly,
- low-eigenvalue modes (*cluster-level structure*) decay slowly,
- the steady-state lies in the Laplacian null space.

5.4 Decay of High-Frequency Components

Let u_k be an eigenvector of L with eigenvalue λ_k. Then its coefficient decays as:

$$c_k(t) = c_k(0)e^{-\gamma\lambda_k t}.$$

Thus:

- if λ_k is large, the component quickly disappears,
- if λ_k is small, the component persists,
- if $\lambda_k = 0$, the component is invariant.

Zeros of the Laplacian correspond to connected components of the graph, i.e., stable semantic basins.

5.5 Steady-State Behavior

As $t \to \infty$:

$$M(t) \to M_\infty,$$

where M_∞ lies in:

$$\ker(L) = \{M : LM = 0\},$$

the space of piecewise-constant vectors on each semantic cluster.

This is the formal mathematical meaning of:

- concept consolidation,
- drift toward stable semantic manifolds,
- loss of noisy or overly specific distinctions.

5.6 Semantic Drift vs. Forgetting

Diffusion is not the same as classical exponential decay:

- **Decay** reduces magnitude globally.
- **Diffusion** removes local irregularities but preserves coarse semantic structure.

Thus diffusion acts as:

- a structural regularizer,
- a semantic smoothing operator,
- a mechanism for turning experience into abstract concepts.

The synergy between decay and diffusion ensures memory remains compact without destroying conceptual organization.

5.7 Stability Properties

Since L is symmetric positive semi-definite:

$$\frac{d}{dt}\|M(t)\|^2 = -2\gamma M^\top LM \leq 0.$$

Hence:

- the system is energy-decreasing,
- memory variance cannot grow,
- the system converges to the smoothest representation consistent with the graph.

This parallels diffusion in heat kernels and manifold learning.

5.8 Role in Artificial Memory Systems

Semantic diffusion supports several desirable behaviors:

1. **Noise reduction**. Sparse, inconsistent, or unstable representations are smoothed.
2. **Cluster strengthening**. Memories with similar content converge toward shared manifolds.
3. **Implicit abstraction**. Fine-grained episodic detail fades, leaving semantic cores.
4. **Prevention of semantic fragmentation**. Diffusion re-aligns drifting representations.
5. **Compatibility with decay**. Diffusion preserves structure even as magnitudes shrink.

5.9 Link to Later Sections

Semantic diffusion represents one component of the broader decay framework. In later sections, diffusion interacts with:

- **linear decay** (global forgetting),
- **competitive inhibition** (global normalization),
- **replay dynamics** (anti-decay consolidation).

Together, these processes maintain a memory system that is stable, compact, and semantically coherent across long time horizons.

6 Replay as Anti-Decay

Overview

Decay gradually reduces the magnitude of memory representations and semantic drift smooths them over the relational graph. If left unopposed, these processes eventually erase all information. Biological memory systems avoid catastrophic loss through *replay*—a mechanism in which previously experienced activation patterns are internally reinstantiated without external input. Replay serves as an "anti-decay" mechanism: it stabilizes important memories, deepens attractor basins, and prevents long-term knowledge from dissipating.

In this section, we develop a mathematically grounded view of replay as a counterforce to decay within a continuous-time dynamical system.

6.1 Replay in Biological Memory Systems

Replay is widely documented in neuroscience, occurring primarily during sleep (especially slow-wave sleep) and brief offline windows during wakefulness. Key empirical properties include:

- **Hippocampal–cortical replay**: Sequences of past experiences reactivate spontaneously, often in compressed time.
- **Consolidation of long-term memory**: Frequently replayed patterns are transferred from hippocampus to neocortex and stabilized into long-term representations.
- **Selective retention**: Memories associated with reward, novelty, or relevance are more likely to be replayed.
- **Noise purification**: Replay removes noise introduced by partial, corrupted, or drifted traces.
- **Synaptic strengthening**: Repetition of an internal pattern increases synaptic efficacy (Hebbian reinforcement).

Replay therefore acts as a targeted stabilizing mechanism rather than a quantitative reversal of decay.

6.2 Replay as Gradient Descent in an Energy Landscape

We model replay as an internal process that pushes memory toward low-energy configurations in the absence of external input. Let $E(M)$ denote an abstract energy functional encoding "memory consistency." Replay is modeled as gradient descent:

$$\frac{dM}{dt} = -\eta \nabla E(M), \qquad I(t) = 0, \qquad \eta > 0.$$

This choice is motivated by theoretical neuroscience and energy-based model formulations. Even without specifying the form of $E(M)$, replay acts as a force that:

- pulls memory toward stable regions,
- sharpens attractors,
- removes drift-induced distortions,
- prevents total decay by reinforcing structure.

6.3 Replay Dynamics Under Linearized Energy Functions

If we approximate the energy around an attractor $M^\star$ using a quadratic expansion:

$$E(M) \approx \frac{1}{2}(M - M^\star)^\top H(M - M^\star),$$

where $H \succeq 0$ is the Hessian, then replay is:

$$\frac{dM}{dt} = -\eta H(M - M^\star).$$

Solution.

$$M(t) = M^\star + e^{-\eta H t}(M(0) - M^\star).$$

Interpretation.

- Large eigenvalues of H correspond to strongly reinforced directions.
- Small eigenvalues correspond to weakly stabilized dimensions.
- Zero eigenvalues define flat directions unaffected by replay.

This captures graded consolidation: some components are strongly preserved, others only weakly.

6.4 Replay vs. Decay: A Balanced System

Decay alone leads to:

$$M(t) \to 0.$$

Replay alone leads to:

$$M(t) \to \arg\min E(M).$$

Together, decay and replay produce a balanced dynamic:

$$\frac{dM}{dt} = -AM - \gamma LM - \delta(1^\top M)1 - \eta\nabla E(M).$$

Replay counteracts decay in a *selective* manner:

- important memories (low-energy regions) are preserved,
- irrelevant memories decay,
- drifting memories are pulled back toward stable attractors,
- noise injected by drift or partial usage is purified.

This yields a memory system that maintains structure while remaining compact.

6.5 Replay as a Filtering Operator

Replay can be interpreted as applying a nonlinear low-pass filter on memory:

$$M(t + \Delta t) = \Phi(M(t)),$$

where Φ moves M closer to low-energy consistent configurations.

Properties:

- **Nonlinear**: depends on the shape of $E(M)$.
- **Selective**: stops short of undoing decay.
- **Conservative**: does not introduce new information.
- **Stabilizing**: ensures important structure is preserved over long horizons.

Replay is therefore not "forgetting in reverse," but noise-removal plus attractor sharpening.

6.6 Replay as a Continuous-Time Alternative to Rehearsal Buffers

Traditional ML approaches experience replay, rehearsal buffers, memory sampling—store raw data and re-feed it to the model. Our formulation differs fundamentally:

- Replay acts on internal memory states, not external data.
- It is continuous-time rather than batch-sampled.
- It performs energy descent instead of gradient ascent.
- It stabilizes existing memory geometry rather than refreshing training data.

This aligns more closely with computational neuroscience than with RL practice.

6.7 Steady States of Replay Dynamics

Let $E(M)$ have minima at $\{M_k^\star\}$. Replay converges toward the locally accessible minimum:

$$\lim_{t\to\infty} M(t) = M_{k^*}^\star.$$

This means:

- replay stabilizes past experiences,
- consolidates clusters,
- deepens existing attractor basins,
- ensures the memory does not collapse under decay.

Replay ensures the system does not converge to the trivial zero-memory solution.

6.8 When Does Replay Trigger?

This paper does not assume any specific triggering mechanism. Replay can be executed:

- during idle periods (sleep-like),
- in background threads,
- periodically at fixed intervals,
- when drift exceeds a threshold,
- when decay has reduced magnitude beyond a limit.

Triggering logic is outside the scope of the theoretical framework, but replay itself—as an operator—is fully characterized.

6.9 Summary

Replay is essential for balancing decay and preserving meaningful structure. Formulated as gradient descent on an internal energy function, replay:

- counteracts excessive decay,
- eliminates noise accumulated from drift,
- sharpens attractors and stabilizes long-term memory,
- preserves semantic organization without requiring external input.

In the next section, we demonstrate how replay interacts with decay and diffusion through synthetic experiments that illustrate long-term stability, cluster coherence, and noise purification.

7 Synthetic Experiments

Overview

This section evaluates the continuous-time decay and replay framework using a series of controlled synthetic experiments. Our goal is not to benchmark against downstream tasks, but to analyze the qualitative and quantitative behavior of the proposed operators:

- linear decay,
- graph Laplacian diffusion,
- differentiated decay rates,
- and replay-based stabilization.

Experiments are intentionally minimalistic to allow precise interpretation of the underlying dynamics.

7.1 Experimental Setup

All simulations model memory as a vector $M(t) \in \mathbb{R}^d$ evolving over time according to a specified ODE. Unless stated otherwise:

- $d = 32$,
- integration uses forward Euler with step size $\Delta t = 0.01$,
- the semantic graph is generated using a Gaussian affinity kernel:

$$W_{ij} = \exp\left(-\frac{\|x_i - x_j\|^2}{2\sigma^2}\right), \qquad \sigma = 1.2,$$

- decay constants use the hierarchy:

$$\alpha_{\text{fast}} = 1.0, \quad \alpha_{\text{mid}} = 0.15, \quad \alpha_{\text{slow}} = 0.02,$$

- diffusion strength is $\gamma = 0.1$,
- replay uses gradient descent with $\eta = 0.25$.

These parameters are chosen to emulate typical relative timescales observed in biological systems.

7.2 Experiment 1: Effect of Linear Decay Alone

We first isolate the effect of pure exponential decay:

$$\frac{dM}{dt} = -AM, \qquad A = \mathrm{diag}(\alpha_1, \ldots, \alpha_d).$$

Protocol.

- A synthetic memory vector $M(0)$ is sampled from $\mathcal{N}(0, I)$.
- Three decay rates are applied: fast, medium, slow.

Results.

- Fast components decay by four orders of magnitude within $t < 3$.
- Medium components retain 20–30% of magnitude over the same period.
- Slow components remain largely unchanged for $t < 5$.

This verifies that diagonal decay enables selective forgetting without destroying stable structure.

7.3 Experiment 2: Semantic Drift via Laplacian Diffusion

Next we isolate diffusion:

$$\frac{dM}{dt} = -\gamma LM.$$

Protocol.

- A noisy memory $M(0)$ is generated by adding Gaussian noise to three prototype cluster vectors.
- Diffusion is applied for $t = 0 \ldots 10$.

Results.

- High-frequency (inter-cluster variance) disappears rapidly.
- Low-frequency (intra-cluster average) remains stable.
- Memory converges to piecewise-constant manifolds determined by semantic clusters.

Spectral analysis confirms:

$$\mathrm{variance}(M(t)) = e^{-2\gamma\Lambda t}\,\mathrm{variance}(M(0)),$$

where large eigenvalue components decay fastest.

7.4 Experiment 3: Combined Decay and Drift

We now evaluate:

$$\frac{dM}{dt} = -AM - \gamma LM.$$

Protocol.

- Initial memories are clustered but noisy.
- Fast, medium, and slow decay rates are applied simultaneously.
- Diffusion is active at $\gamma = 0.1$.

Findings.

- Short-term noise decays quickly due to large α_{fast}.
- Medium-scale features drift toward cluster centroids.
- Slow-scale semantic features remain intact.

This demonstrates the system's natural tendency to compress memory while preserving core semantics.

7.5 Experiment 4: Replay as Anti-Decay

We add replay:

$$\frac{dM}{dt} = -AM - \gamma LM - \eta \nabla E(M).$$

For these experiments, the energy function $E(M)$ is chosen as a mixture of quadratic wells centered at three prototype memories:

$$E(M) = \sum_{k=1}^{3} \frac{\omega_k}{2} \|M - M_k^\star\|^2, \qquad \omega_k > 0.$$

Protocol.

- Memory is initialized near—but not inside—one of the wells.
- Decay and diffusion push the memory away from the prototype.
- Replay pulls the memory back toward the prototype.

Results.

- Without replay: memory collapses toward zero.
- With replay: memory converges to the nearest prototype.
- Drift-induced distortions are eliminated.

Replay restores long-term stability that decay and diffusion alone would remove.

7.6 Experiment 5: Competition-Driven Normalization

We evaluate the effect of the global competition term:

$$\frac{dM}{dt} = -\delta(1^\top M)\,1.$$

Protocol.

- Memories with artificially inflated magnitude are created.
- Competition is applied with $\delta = 0.05$.

Results.

- All components move toward a scale-normalized equilibrium.
- No direction-specific information is lost.
- Excess energy is removed uniformly across memory dimensions.

Competition prevents runaway activation induced by repeated stimuli.

7.7 Experiment 6: Combined System Dynamics

Finally, we evaluate the full decay–drift–replay system:

$$\frac{dM}{dt} = -AM - \gamma LM - \delta(1^\top M)1 - \eta\nabla E(M).$$

Protocol.

- Initial memory vectors contain fine-grained episodic noise on top of stable cluster-level structure.
- The system is simulated for $t = 0 \dots 20$.

Findings.

- Episodic noise is eliminated quickly (fast decay).
- Cluster-level variance is smoothed (Laplacian diffusion).
- Global magnitude stays bounded (competition).
- Stable attractor structure persists indefinitely (replay).

The interplay of decay, drift, and replay produces a memory system that is:

- compact,
- coherent,
- stable,
- and resistant to catastrophic forgetting.

7.8 Summary of Experimental Insights

Across all simulations, several key principles emerge:

1. **Decay** removes unused or unstable components rapidly.
2. **Diffusion** reorganizes memory into smooth semantic manifolds.
3. **Differentiated decay rates** preserve long-term knowledge.
4. **Replay** counteracts drift and stabilizes attractors.
5. **Competition** ensures global scale normalization.
6. The combined system achieves **long-term stability without unbounded memory growth**.

These experiments collectively validate the theoretical framework presented in the earlier sections and demonstrate how decay and replay cooperate to maintain a coherent memory state over time.

8 Discussion

Overview

The experiments and theoretical results converge toward a consistent picture: forgetting is not a singular operation but an emergent property of multiple cooperating decay processes. This section discusses how the proposed continuous-time framework unifies biological and computational perspectives on memory, and how differentiated decay, diffusion, and replay interact to yield long-term stability in artificial systems.

8.1 Interplay Between Decay, Drift, and Replay

The central insight emerging from this work is that memory stability arises not from preventing decay but from *balancing* decay with structure-preserving forces.

We observe three fundamental processes:

1. **Linear decay** reduces the magnitude of memory content.
2. **Semantic diffusion** smooths and reorganizes that content within a relational graph.
3. **Replay** counteracts drift by returning the system toward stable internal configurations.

Each process alone is insufficient:

- Decay alone erases all structure.
- Diffusion alone homogenizes memory without discrimination.
- Replay alone amplifies noise and prevents abstraction.

When combined, however, they produce:

- compact representations,
- stable semantic organization,
- resistance to catastrophic forgetting,
- graceful degradation of unused content,
- and long-term retention of relevant knowledge.

This balance closely parallels biological systems, where decay, drift, and replay operate simultaneously across multiple scales.

8.2 The Role of Differentiated Decay Rates

A key conceptual contribution of this work is the formalization of *multi-timescale forgetting.* Biological memory operates across diverse temporal layers—rapid synaptic decay, medium-scale cortical smoothing, and slow structural reorganization.

Our framework captures this by assigning distinct decay constants to different memory dimensions:

$$\alpha_{\text{fast}} \gg \alpha_{\text{mid}} \gg \alpha_{\text{slow}}.$$

This hierarchy produces the following emergent behaviors:

- Fast-decaying components represent working memory.
- Medium-decaying components form intermediate concepts.
- Slowly decaying components encode long-term semantic structure.

Differentiated decay is therefore a principled way to impose *temporal abstraction*—a core requirement for long-lived AI systems.

8.3 Semantic Drift as a Means of Compression

Drift is often viewed negatively in artificial memory systems, but our results suggest a more nuanced interpretation. Semantic diffusion:

- reduces high-frequency noise,
- merges redundant information,
- amplifies cluster-level structure,
- and smooths representations into low-dimensional manifolds.

From this perspective, drift acts as an implicit form of:

- dimensionality reduction,
- semantic abstraction,
- and memory compression.

This aligns with long-standing theories in both neuroscience and cognitive psychology, where forgetting is often a precondition for the formation of generalizable concepts.

8.4 Replay as a Stabilization Mechanism

Replay emerges as the counterweight to decay and drift. By moving memory toward low-energy configurations, replay selectively enhances stable components of memory while ignoring transient noise.

Replay's contribution is twofold:

1. **Stability:** It anchors memory to stable internal attractor-like patterns.
2. **Purification:** It removes drift-induced distortions and reinforces consistency.

Thus, replay prevents the trivial solution $M(t) \to 0$ and preserves structured memory over long timescales, even in the presence of continuous decay.

8.5 Biological Interpretability of the Framework

The proposed mathematical operators align well with biological mechanisms:

- Linear decay captures synaptic weakening (LTD) and spontaneous neuronal fading.
- Diffusion reflects cortical smoothing, association strengthening, and manifold-level abstraction.
- Replay corresponds to hippocampal–cortical consolidation and offline associative reinstantiation.
- Differentiated timescales match the known hierarchy of memory systems from sensory buffers to long-term semantic knowledge.

This biological plausibility suggests that continuous-time decay and replay models may provide a foundation for more human-like long-term memory systems in artificial agents.

8.6 Implications for Long-Term AI Systems

Agentic systems deployed over long horizons require memory systems that:

- remain stable without growing unbounded,
- forget irrelevant or outdated information,
- consolidate meaningful patterns,
- adapt to changing environments,
- and preserve long-term trends and user-specific knowledge.

The proposed framework supports these requirements naturally. In particular:

- **Capacity-limited agents**: Differentiated decay prevents unbounded memory accumulation.
- **Contextual personalization**: Replay preserves stable user-specific patterns.
- **Continual learning**: Diffusion smooths transitions and prevents fragmentation.
- **Non-catastrophic forgetting**: Slow decay ensures semantic structures persist despite continuous updates.

8.7 Beyond Storage: Memory as a Dynamical Substrate

The broader conceptual takeaway is that memory should be treated not as a database or static repository but as a *dynamical substrate*. Continuous-time evolution enables:

- emergence of attractor-like representations,
- graceful forgetting,
- compression through drift,
- stability via replay,
- and temporal abstraction via timescale separation.

This perspective shifts the focus away from static storage toward dynamic self-organization—an essential step for building memory systems capable of long-term reasoning and adaptive behavior.

8.8 Synthesis

Taken together, the results demonstrate that:

1. forgetting is a structured process with multiple interacting components,
2. diffusion and decay operate as complementary forms of compression,
3. replay provides targeted reinforcement that prevents information collapse,
4. and multi-timescale modeling is critical for intelligent systems that must operate continuously over extended periods.

The theory presented here thus provides a principled foundation for artificial memory systems that are compact, stable, and biologically inspired.

9 Limitations

Overview

While the continuous-time framework for memory decay, drift, and replay presented in this work is broadly applicable and biologically inspired, it also comes with a number of conceptual, computational, and modeling limitations. These limitations highlight important directions for future research and clarify the scope of the current work.

9.1 Simplifying Assumptions in the Mathematical Model

The formulation relies on several mathematical assumptions that, while standard in theoretical analysis, may oversimplify real-world memory dynamics.

Linearity of decay operators. We assume exponential decay generated by linear operators such as AM and LM. Biological synapses often exhibit non-linear decay rules depending on firing rates, homeostatic constraints, or molecular signaling pathways. Linear decay provides tractability but does not capture these complexities.

Diffusion governed strictly by the Laplacian. Laplacian diffusion assumes symmetric relations and smoothness across the semantic graph. However, semantic change in large systems can be non-local, asymmetric, or state-dependent. Diffusion captures one important mode of semantic drift but not all.

Quadratic energy approximation for replay. Replay is modeled through gradient descent on an energy functional, often approximated quadratically near local minima. In complex cognitive systems, replay may rely on non-linear reinstatement dynamics that cannot be expressed as simple energy descent.

9.2 Dependence on Graph Quality

The accuracy of semantic diffusion depends heavily on the quality of the underlying graph structure.

Noisy or misaligned graphs. If the affinity matrix W does not accurately reflect semantic relations, diffusion can degrade meaningful distinctions or merge unrelated memories.

Static graph assumption. The framework treats the graph as fixed during the experiments. In practice, semantic structure evolves over time and may require adaptive or self-organized graph updates.

Sensitivity to kernel choice. Gaussian affinity kernels, $k(x_i, x_j) = e^{-\|x_i - x_j\|^2/(2\sigma^2)}$, are convenient but may not represent the true geometry of high-dimensional embeddings.

9.3 Abstract Treatment of Replay

Although replay is modeled as an internal stabilization mechanism, several limitations remain:

No mechanism for sequence replay. Biological replay often reinstates temporal sequences of states (e.g., hippocampal place-cell cascades). The presented model captures only static pattern reinstantiation, not temporal structure.

No selective prioritization signal. Empirical studies show that reward, novelty, and surprise strongly bias what gets replayed. Our formulation treats replay uniformly without incorporating prioritization signals.

Energy-based formulation abstracts away circuit-level details. The gradient-descent interpretation provides mathematical clarity but does not model rhythmic oscillations (theta/gamma), inhibitory gating, or hippocampal–cortical interactions known to drive replay.

9.4 Limited Connection to Downstream Tasks

The focus of this work is the internal dynamics of memory states themselves. We do not evaluate:

- retrieval accuracy,
- downstream reasoning performance,
- long-horizon predicting or planning reliability,
- integration with large language models or multimodal agents.

This isolation is intentional for theoretical clarity but limits direct empirical comparability with existing machine-learning memory architectures.

9.5 Scalability Considerations

Although the theoretical framework is lightweight, certain components raise scalability concerns when implemented in large systems.

Laplacian multiplication. Computing LM requires $O(|E|)$ operations. For dense graphs this becomes $O(d^2)$, which may be prohibitive for large memory dimensions.

Spectral interpretations require eigen decompositions. While eigen decompositions are used only in analysis (not in the system itself), scaling to tens of thousands of memory dimensions is non-trivial.

Replay frequency. If replay is triggered too often, it may introduce computational overhead. If triggered too rarely, drift may become unbounded.

9.6 Absence of Ground-Truth Biological Validation

Although the model is informed by well-established neuroscientific principles, it does not claim biological fidelity. Specifically:

- synaptic plasticity is highly nonlinear and stochastic,
- cortical consolidation unfolds across multiple interacting regions,
- real neural manifolds are embedded in recurrent networks, not linear ODEs.

Our framework captures high-level qualitative behaviors but does not attempt to model biological circuits in detail.

9.7 Lack of Adaptive Parameter Dynamics

Decay rates α_i, diffusion strength γ, and replay step size η are treated as fixed constants. In practice, these parameters may need to:

- adapt over time,
- depend on memory usage statistics,
- respond to environmental volatility,
- or vary across different memory modalities.

Static parameters simplify analysis but constrain the generality of the model.

9.8 Limited Exploration of Noise and External Perturbations

The current experiments assume clean dynamical evolution without:

- stochastic drift,
- environmental noise,
- adversarial interference,
- or competing tasks.

Real-world memory systems must remain stable under such conditions.

9.9 Summary

In summary, the proposed decay–drift–replay framework:

- offers a clean mathematical foundation for continuous forgetting,
- captures many known principles of biological memory,
- and demonstrates desirable stability properties,

but also relies on simplifying assumptions and idealized conditions. Addressing these limitations—particularly adaptive parameter dynamics, realistic noise models, selective replay, and scalable graph structures—remains important future work.

10 Conclusion

Conclusion

This work develops a continuous-time theoretical framework for understanding forgetting, semantic drift, and memory stabilization in artificial systems. By decomposing memory evolution into three interacting processes—linear decay, graph-based diffusion, and replay-driven reinforcement—we show that forgetting is not merely a destructive force but a structural component of long-term memory organization.

Our analysis demonstrates several central points:

1. **Differentiated decay** provides a principled mechanism for modeling multi-timescale memory phenomena such as short-term fading, intermediate consolidation, and slow semantic erosion.

2. **Semantic diffusion** smooths memory across relational structure, removes high-frequency noise, and yields coherent low-dimensional manifolds that reflect stable conceptual organization.

3. **Replay** acts as an internal anti-decay process, pulling drifted states back toward stable configurations and preventing catastrophic forgetting in the presence of continuous decay.

4. **The combined dynamics** produce compact, stable, and semantically meaningful memory representations without requiring bounded buffers or ad hoc pruning mechanisms.

Together, these insights support the broader thesis that memory in long-lived artificial agents should not be treated as a static store, but as a *dynamical substrate*—one that evolves continuously through the interplay of forgetting, drift, and stabilization. This view aligns closely with biological memory systems, where transient activations, synaptic decay, semantic smoothing, and offline replay interact to produce stable, long-term knowledge.

While the present work focuses on the mathematical foundations of this continuous-time framework, it opens several promising directions for future research:

- integrating adaptive decay rates that depend on usage patterns,
- incorporating selective replay based on reward, novelty, or uncertainty,
- developing scalable graph structures that evolve over time,
- coupling the dynamical memory model with large language models or agent architectures,
- and exploring the relationship between dynamical memory and long-horizon reasoning or planning.

Ultimately, understanding and formalizing forgetting is essential for building artificial systems that remain stable, compact, and adaptable over long operational lifetimes. We hope that this work provides a clear and principled foundation for future efforts toward memory systems that, like biological counterparts, continually organize, abstract, and reinterpret their internal states over time.

References

György Buzsáki. Hippocampal sharp wave–ripple: A cognitive biomarker for episodic memory and planning. *Hippocampus*, 25(10):1073–1188, 2015.

Fan R. K. Chung. *Spectral Graph Theory*. American Mathematical Society, 1997.

Peter Dayan and Laurence F Abbott. *Theoretical Neuroscience: Computational and Mathematical Modeling of Neural Systems*. MIT Press, 2001.

Donald O Hebb. *The Organization of Behavior: A Neuropsychological Theory.* Wiley, 1949.

John J Hopfield. Neural networks and physical systems with emergent collective computational abilities. *Proceedings of the National Academy of Sciences*, 79(8):2554–2558, 1982.

Dmitry Krotov and John J Hopfield. Dense associative memory for pattern recognition. *Advances in Neural Information Processing Systems*, 29, 2016.

Hubert Ramsauer, Bernhard Schäfl, Johannes Lehner, Pascal Seidl, Michael Widrich, Thomas Adler, Lukas Gruber, Markus Holzleitner, Miruna Pavlović, Geir Kjetil Sandve, Victor V Greiff, Sepp Hochreiter, and Günter Klambauer. Hopfield networks is all you need. *Advances in Neural Information Processing Systems*, 33, 2020.